Necessity Entrepreneurs,*

David Chivers† Zhigang Feng‡ Elisa Keller§ Bo Li¶

February 18, 2026

Abstract

We find...

JEL Classification: E23, I10, O40.

Keywords:Occupational Choice, Entrepreneur, Misallocation,

*We thank

†Department of Economics and Finance, University of Durham, Durham, DH1 3LB, david.chivers@durham.ac.uk

‡Department of Economics, University of Nebraska, Omaha, NE 68106, United States, z.feng2@gmail.com.

§Department of Economics, University of Exeter, elisa.keller@exeter.ac.uk

¶Department of Economics, Peking University, lb.harbin@gmail.com .

1 Introduction

Entrepreneurship, often associated with innovation and firm growth, is a cornerstone of economic development. Yet, this image predominantly applies to

untries, where a higher share of entrepreneurs report starting firms out of necessity. When considering only those who report starting a business for opportunistic reasons, the relationship between entrepreneurship and economic development becomes positive (Chivers, 2017).

Defining opportunity-driven versus necessity-driven entrepreneurship, however, presents several challenges. The Global Entrepreneurship Monitor (GEM) provides nationally representative survey data and classifies entrepreneurs as either opportunity-driven or necessity-driven using the question: "Are you involved in this start-up to take advantage of a business opportunity or because you have no better choices for work?" Naturally, relying on subjective, self-reported data introduces potential measurement error. In contrast, Fairlie and Fossen (2019) propose an empirical approach in which opportunity-driven entrepreneurs are defined simply as individuals who were previously employed, while necessity entrepreneurs are defined as those who were previously unemployed. Although this definition is empirically tractable, it cannot fully capture the underlying motivation: for example, someone claiming unemployment insurance may claim to be seeking work but is actually working on a business thus appear as a "necessity" entrepreneur, or someone employed may start a firm to hedge against potential job loss and appear an "oppotunistic" enterpreneur.

In this paper, we provide a theoretical definition of entrepreneurship based on agents' optimal occupational choices, taking into account their managerial talent, employment status, educaiton and wealth(?) We define opportunity-driven entrepreneurs as those who would choose self-employment regardless of their employment circumstances, and necessity entrepreneurs as those who would not have chosen entrepreneurship if other employment options were available. This conceptualization allows us to frame necessity entrepreneurship as a potential form of misallocation, where individuals engage in entrepreneruship not because it is their most productive option, but because external constraints limit their alternatives. We calibrate are model to the US economy and find...

This paper is related to the following

2 The Model: Economic Environment

Consider a Lucas (1978) span of control model, where individuals differ in the level of human capital, h. As in Gomez and Kuhen (2017), individuals are educated to either the primary (P), secondary (S) or tertiary level (T). The level of education means that a tertiary educated individuals have a higher draw of human capital, on average, than secondary educated individuals, who in turn have a higher draw on average than primary educated individuals. This human capital draw is positively correlated with managerial ability x as well as labor productivity z.

At the start of the period agents enter as either unemployed ($i_w = 0$), employed ($i_w = 1$), or an entrepreneur ($i_w = 2$). Individuals face a choice of whether to become an entrepreneur, to continue working or to search for work. However, the value of employment is different depending on whether an agent enters the period as employed ($i_w = 1$) or not. Although there is a risk that an employee may become separated from a firm with probability ρ, this is less risky than searching for work which has a $p(\theta)$ probability of obtaining employment. Consequently, the expected value of employment is higher for those already in employment than those who are not. Additionally, the probability of separation is lower for individuals with higher levels of human capital and they are able to command a higher wage due to higher levels of productivity. So the value of employment, holding other things equal, is higher for those with higher human capital as well.

The innovation of this model is that the entrepreneurial decision results in two types of entrepreneurs emerge, opportunity-based entrepreneurs and necessity-based entrepreneurs. We define opportunity-based entrepreneurs are those whose value of entrepreneurship is higher, even if they had employment. Necessity-based entrepreneurs are those who would not have become entrepreneurs, if they had employment.

Timing: The timing of events can be summarized below. Occupational choice happens at the start of time t. Those choosing to become entrepreneurs run projects and hire workers on on rolling contracts (i.e. contracts, will be reassessed each period). Those who were previously employed within a project can decide to carry on working but carry a risk of separation with probability ρ. If separation occurs these workers will claim unemployed benefits and be unable to look for work until the following period. After separation occurs, vacancies are posted and who were unemployed in the previous period are matched with firms. Those that are successful and match with with probability $p(\theta)$ go on to earn a wage. Whereas those that are unsuccessful will claim unemployment benefits and be unable to search again until the following period.

Unlike the search and matching literature there is no wage bargaining between workers and the firm at the end of matching. Instead the wage pins down the occupational choice decision which determines those wanting to become entrepreneurs and those who will search for work. Crucially, the worker only finds out whether they enter production and earn a wage after their occupational choice decision.

Figure 1: Timing for Matched Workers

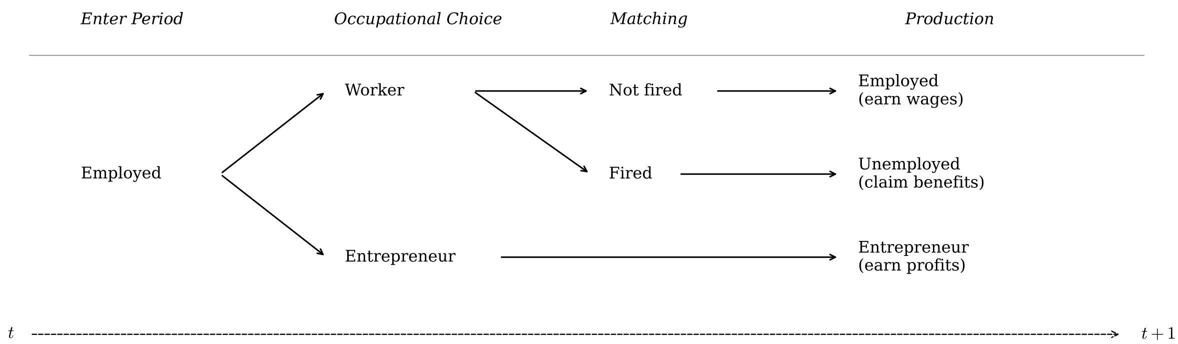
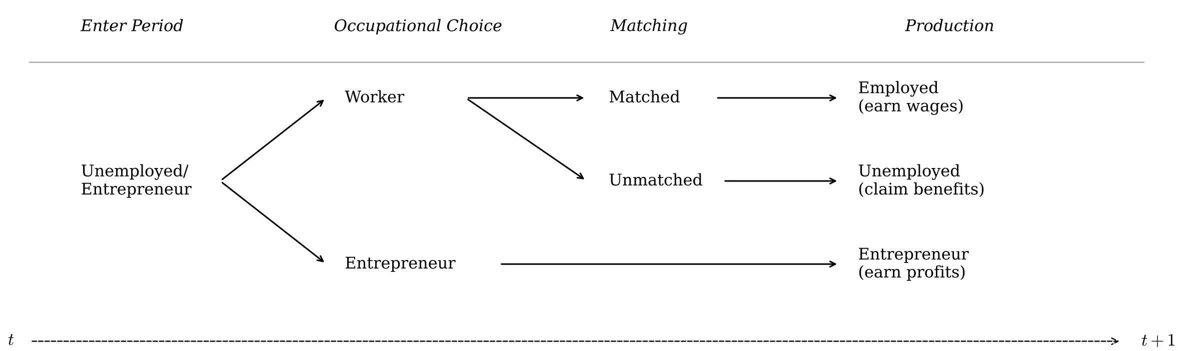


Figure 2: Timing for Unmatched Workers



2.1 Preferences, endowments and technology

Preferences: Consumption by an agent in period t is c_t, with utility given by $U(c_t)$.

Endowments: Agents are endowed with an initial capital asset, a_0, which can be used as an input in production. Each individual is endowed with human capital shock h^i. Human capital h is positively correlated with managerial talent x, labour productivity z and seperation rate ρ.

Production: Firms use labor n and capital k to produce a single consumption good, y. Capital depreciates at a constant rate of δ. Managers can operate only one project. The functional form of the production function is:

$$y = xk^{\alpha}(zn)^{\gamma} \text{ where } \alpha, \gamma > 0. \tag{1}$$

Managerial talent x is determined by the Pareto distribution $F(x) = \left[1 + \frac{x-\varsigma}{\xi}\right]^{-\vartheta_i}$ where ϑ is the shape parameter, ς the location parameter and ξ is the scale parameter.

Factor remuneration: Firms rent capital at the common market rate r.

Government: The government runs a balanced budget each period and provides unemployment insurance which is financed through lump sum taxation τ, which is shared by the employee $(1-\psi)\tau$ and employer $\psi\tau$. This program guarantees each household an unemployment benefit of b.

2.2 The Firm's problem

The firm's problem is:

$$\max_{n,k} Xk^{\alpha}n^{\gamma} - (1+\tau\psi)wzn - rk - \kappa\frac{\rho zn}{q(\theta_t)} \tag{2}$$

employees must pay a share of unemployment insurance τ to all workers who produce output. Firms have to pay search costs κ for all vacancies posted. The number of vacancies posted is designed to replace any lost workers, ρzn. This, however, is affected by the firm-hiring rate $q(\theta_t) = \bar{m}\left(\frac{v_t}{u_t}\right)^{-\mu}$, where aggregate vacancies v_t is simply the aggregate level of all those who have separate from firms $\sum \rho zn$ within the period and u_t is the total unemployed looking for work within that period (see appendix).

When the market is tight and the hiring rate is low, firms have to search more and thus face higher search costs in order to replace lost workers. This is because $0 < q(\theta_t) < 1$,so a lower $q(\theta_t)$ increases $\frac{\rho zn}{q(\theta_t)}$. If the firm lose 10 workers, they want to hire 10 workers to replace them. But if the hiring rate, $q(\theta_t)$, is 50% then they will face double the amount of search costs than if $q(\theta_t)$ were 100%. The upshot of this is that when markets are tight, the size of the desired firm size n decreases ex ante , as the cost of replacing workers is high and so entrepeneurs would prefer to run smaller firms.

To avoid issues of directed search, we assume that employees are randomly matched with firms (as in Chivers et al., 2015) and each firm employees a distribution of workers accross the human capital spectrum such that each firm employs the market averaege.

3 Optimal behavior and equilibrium

In any time period t agents are distinguished by (h, a, x, i, z). The timing of the economy is given as follows.

1. Households enter each new period with assets a and an employment status i_w
2. Idiosyncratic shocks h is drawn which determines x (managerial ability) and z (labor productivity), ρ seperation rate conditional on education level, P, S or T.
3. Households make an occupation decision: entrepreneur ($\mathcal{I}_e = 1$) or worker ($\mathcal{I}_e = 0$).
4. Capital and labor markets clear. A fraction ρ of employees separate from firms and $p(\theta)$ are hired to replace them
5. Households choose: consumption (c), borrowing/saving (a')

3.1 Firm manager

Firms are distinguished by their productivity realization x. Agents with sufficient ability to become managers choose the level of capital and the number of employees to maximize profit subject to a technological constraint. In order to simplify the exposition, first consider the problem of a manager with talent x^i for a given level of capital k (i.e., labor input choice only):

$$\max_{zn} xk^{\alpha}(zn)^{\gamma} - zw(1+\psi\tau)n - rk - \kappa\frac{\rho zn}{q(\theta)} \tag{3}$$

The first order conditions are:

$$\gamma xk^{\alpha}zn^{\gamma-1} - w(1+\psi\tau) - rk - \frac{\rho\kappa}{q(\theta)} \tag{4}$$

$$n^*(k, x, \widetilde{w}) = \left[\frac{\gamma xk}{(w(1+\psi\tau) + \frac{\rho\kappa}{q(\theta)}}\right]^{\frac{1}{1-\gamma}} \tag{5}$$

Substituting (5) into (3) yields the manager's profit function for a given level of capital:

$$y(k,x,\widetilde{w}) = xk\left[\frac{\gamma x k^{\alpha}}{(w(1+\psi\tau)+\frac{\rho\kappa}{q(\theta)}}\right]^{\frac{\gamma}{1-\gamma}} \tag{6}$$

3.1.1 Capital

Initially we assume capital markets function perfectly.

3.2 Entrepreneurs

Entrepreneurs maximize expected discounted utility of consumption

$$\max_{\{c_t,a_{t+1}\}} \mathbb{E}\sum_{t=0}^{\infty}\beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1+r-\delta)a + \nu - \tau_{\pi} \tag{7}$$

$$a' \geq -\bar{a}$$

3.3 Employed

The workforce (previous periods employed who choose to reapply for work) maximize expected discounted utility of consumption

$$\max_{\{c_t,a_{t+1}\}} \mathbb{E}\sum_{t=0}^{\infty}\beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1+r-\delta)a + (1-\rho)\left(wz(1-(1-\psi)\tau_y)\right) + \rho b \tag{8}$$

$$a' \geq -\bar{a}$$

where b is the unemployment benefit.

3.4 Unemployed

The unemployed maximize expected discounted utility of consumption

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1 + r - \delta)a + p(\theta)\left(wz(1 - (1 - \psi)\tau_y)\right) + (1 - p(\theta))b \tag{9}$$

$$a' \geq -\bar{a}$$

where b is the unemployment benefit.

3.5 Government

The government runs a balanced budget with a lump-sum taxes $T(y) = (y - \lambda_p y^{1-\tau_p})$

3.6 The household's problem

Let i_w denote the household's labor market status in the previous period, with $i_w = 2$ indicating being entrepreneur, $i_w = 1$ being employed and $i_w = 0$ being unemployed. Let $\mathcal{I}_e$ indicate occupational choice, where $\mathcal{I}_e = 1$ if the household is an entrepreneur and $\mathcal{I}_e = 0$ if the household is a worker.

Let denote $\mathbb{E}_\rho$ the expected value considering the possibility of being separated from the job as employed; $\mathbb{E}_{p(\theta)}$ the expected value considering the possibility of finding new job if previously unmatched with a firm (unemployed former entrepreneurs). V_w is the contemporary utility as a worker, which varies with the employment status. $\iota_{i_w=1}$ is an indicator function. We can write the household's problem recursively as follows. More specifically,

$$\begin{aligned}\mathbb{E}_{p(\theta)}\left(V_w + \beta\mathbb{E}\mathbf{V}(a', x', i_w')\right) = p(\theta)\left(V_n + \beta\mathbb{E}\mathbf{V}(a', x', i_w' = 1)\right) \\ + (1 - p(\theta))\left(V_u + \beta\mathbb{E}\mathbf{V}(a', x', i_w' = 0)\right)\end{aligned}$$

$$\begin{aligned}\mathbb{E}_\rho\left(V_w + \beta\mathbb{E}\mathbf{V}(a', x', i_w')\right) = \rho\left(V_u + \beta\mathbb{E}\mathbf{V}(a', x', i_w' = 0)\right) \\ + (1 - \rho)\left(V_n + \beta\mathbb{E}\mathbf{V}(a', x', i_w' = 1)\right)\end{aligned}$$

Notice that for agents who recently separated from the firm cannot rematch with a new firm in the current period.

$$\mathbf{V}(a,h,i_w)=\max_{\{a',c,p(\theta),\mathcal{I}_e,\mathcal{I}_{\mathrm{w}}\}}\left\{\begin{array}{c}\iota_{i_w=0}\left[(1-\mathcal{I}_e)\mathbb{E}_{p(\theta)}\left(V_w+\beta\mathbb{E}\mathbf{V}(a',x',i_w')\right)+\mathcal{I}_e\left(V_e+\beta\mathbb{E}\mathbf{V}(a',x',i_w'=0)\right)\right]+\\ \iota_{i_w=1}\left[(1-\mathcal{I}_e)\mathbb{E}_{\rho}\left(V_w+\beta\mathbb{E}\mathbf{V}(a',x',i_w')\right)+\mathcal{I}_e\left(V_e+\beta\mathbb{E}\mathbf{V}(a',x',i_w'=0)\right)\right]+\\ \iota_{i_w=2}\left[(1-\mathcal{I}_e)\mathbb{E}_{p(\theta)}\left(V_w+\beta\mathbb{E}\mathbf{V}(a',x',i_w')\right)+\mathcal{I}_e\left(V_e+\beta\mathbb{E}\mathbf{V}(a',x',i_w'=0)\right)\right]\end{array}\right\}$$

subject to

$$c+a'\leq(1-\tilde{r}-\delta)a+inc \tag{10}$$

$$inc=\begin{cases}ra+\widetilde{w}_n & \text{if } \mathcal{I}_e=0,\ i_w=1\\ ra+\widetilde{w}_u & \text{if } \mathcal{I}_e=0, i_w=0\\ ra+\nu(k,x;\tilde{r},w,\tau) & \text{if } \mathcal{I}_e=1\end{cases}$$

where inc is the earnings of the worker or entrepreneur, $\widetilde{w}_n=(1-\rho)\left\{zw(1-(1-\psi)\tau)\right\}+\rho b$ and $\widetilde{w}_u=p(\theta)\left\{zw(1-(1-\psi)\tau)\right\}+\{1-p(\theta)\}b$.

3.7 Equilibrium

Let $\hat{h}$ be the critical value of human capital that determines occupational choice which is affected by both employment status and education level. If $h\geq\hat{h}$ then individuals choose entrepreneurship over becoming a worker. The critical value of human capital is always higher for those who have employment than those who are unmatched with a firm $\hat{h}_{i_w\neq1}<\hat{h}_{i_w=1}$. We define necessity entrepreneurship as those who would not have become an entrepreneur *if* they had an employment contract:

$$\hat{h}_{i_w\neq1}<h<\hat{h}_{i_w=1}.$$

We define opportunity-driven entrepreneurship as those who would become an entrepreneur *regardless* of circumstance:

$$h\geq\hat{h}_{i_w=1}\geq\hat{h}_{i_w\neq1.}$$

As there are three education levels, there are a total of six critical values of human capital: two for primary educated matched or unmatched, $\hat{h}^P_{i_w=1}$and $\hat{h}^P_{i_w\neq1}$ two for secondary educated,$\hat{h}^S_{i_w=1}$and $\hat{h}^S_{i_w\neq1}$, and two fo

4 Calibration

Table 1: Parameter values, baseline economy

Parameters	Values	Description	Comments/observations/target
β	0.94	Discount factor	
σ	2.0	Risk aversion	$u(c) = \frac{c^{1-\sigma}}{1-\sigma}$
δ	0.06	Capital depreciation	
α	0.3207	Capital share	
γ	0.4693	Labour Share	Frac. of entrepreneurs
h	$\{0.109, 0.581, 0.310\}$	Share of population by h	Barro and Lee (2013)
ϑ_e	$\{0.967, 1.007, 1.156\}$	Shape parameter by h	Mean firm size, Gomez and Kuhen (2017)
ς	2.02655	Location parameter	$F(x) = \left[1 + \frac{x-\varsigma}{\xi}\right]^{-\alpha_i}$
ξ	0.1115	Scale parameter	Gomez and Kuhen (2017)
z	$\{1.0, 1.31, 1.61\}$	Wage premium by h	Goldin and Katz (2009)
ρ	$\{0.0383, 0.0288, 0.0136\}$	Job separation rate by h	Cairo and Cajner (2016)
$\bar{m}$	0.8	Matching efficiency	Lubik, 2013 $p(\theta) = \bar{m}\left(\frac{v}{u}\right)^{1-\mu}$
μ	$[0.5, 0.7]$	Matching elasticity	Petrongolo and Pissarides (2001)
κ	$[0.03, 0.045] \cdot w$	Search costs	(Hagedorn and Manovskii, 2008)
τ_π	0	Corporation tax	Set as zero for now
τ_y	0.151	Income Tax	$T(y) = (y - \lambda_p y^{1-\tau_p})$
λ_p		Income Tax	Determined in equm.
b	$0.40 \cdot w,$	Social security benefits	Shimmer and Krause and Lubik (2005)
ψ	0	Share	
Δ	0	Credit spread	
$\bar{a}$	0	Borrowing constraint	

5 Baseline Results

6 Quantitative Analysis

6.1 No Unemployment

This experiment is designed to eliminate necessity entreprenueurship. As there is no risk of becoming unemployed, the value of entrepenuership increases at all education levels leading to more ntreprenueurs with high firm sizes.

6.2 Increasing Unemployment Insurance (Done)

While holding taxes constant (Done) Increase unemployment insurance benefit b while keeping the taxation level constant. Since the insurance benefit is funded through taxes, this

Table 2: Impact of unemployment on entrepreneurship

	Baseline	no Unemployment
entrepreneur-avg	0.145	0.197
entrepreneur-low	0.143	0.207
entrepreneur-mid	0.137	0.184
entrepreneur-high	0.161	0.216
firm size-avg	21.877	24.126
firm size-low	9.682	9.884
firm size-mid	18.007	19.814
firm size-high	33.417	37.214
Unemp-Ent	0.00204	0

Table 3: Impact of UI on entrepreneurship

	Baseline, UI=0.4	UI=0.3	UI=0.0	UI=0.6
entrepreneur-avg	0.145	0.145	0.193	0.144
entrepreneur-low	0.143	0.141	0.197	0.135
entrepreneur-mid	0.137	0.136	0.187	0.141
entrepreneur-high	0.161	0.164	0.181	0.154
firm size-avg	21.877	21.550	14.68	21.177
firm size-low	9.682	9.765	6.13	9.814
firm size-mid	18.007	17.931	11.33	17.023
firm size-high	33.417	32.477	23.98	32.958
Unemp-Ent	0.00204	0.00197	0.005	0.00084

change will result in an additional government surplus. This surplus will then be distributed back to the population as a lump-sum transfer. [DC: This does make sense? it is equivilant of lowering taxes? When tax is larger than benefits there will be a surplus, when benefits is larger than tax there will be a deficit. The goverment should keep the surplus and borrow at zero interest the deficit as it is just a hypothetical.]

While adjusting taxes adjust We will adjust the insurance benefit as previously described. However, we will also modify the tax rate to ensure that the reform is revenue-neutral. [DC: Has this been done?]

6.3 Consumption floor

Lump-sum transfer: In this experiment, we will introduce a consumption floor, denoted as $\underline{c}$, ensuring that households receive a lump-sum transfer when their disposable income falls below a certain threshold. This transfer is designed to guarantee a minimum consumption level of $\underline{c}$ for each household. The funding for this consumption floor will be sourced from general tax revenues.

using ui = 0.1 as baseline.

Table 4: Impact of consumption floor on entrepreneurship, UI=0.4

	Baseline, UI=0.4	Lump-sum transfer	Penalty
entrepreneur-avg	0.145	0.144	0.144
entrepreneur-low	0.143	0.143	0.142
entrepreneur-mid	0.137	0.136	0.137
entrepreneur-high	0.161	0.160	0.159
firm size-avg	21.877	21.867	21.951
firm size-low	9.682	9.655	9.776
firm size-mid	18.007	17.988	17.923
firm size-high	33.417	33.431	33.781
Unemp-Ent	0.00204	0.00189	0.00177

Table 5: Impact of consumption floor on entrepreneurship, UI=0.0

	Baseline, UI=0.4	Baseline, UI=0.0	Penalty
entrepreneur-avg	0.145		
entrepreneur-low	0.143		
entrepreneur-mid	0.137		
entrepreneur-high	0.161		
firm size-avg	21.877		
firm size-low	9.682		
firm size-mid	18.007		
firm size-high	33.417		
Unemp-Ent	0.00204		

see how entrepreneurship changes with UI and $\underline{C}$. want to see the case when $\underline{C}$ binds, also to report statistics on saving.

Non-pecuniary penalty: Alternatively, the households will face a non-pecuniary utility penalty if their consumption falls below $\underline{c}$. This is to reflect the fact of no existence of social protection in most developing countries.

6.4 Consumption floor with no unemployment or taxes

6.5 Credit friction wedge

Add a wedge on borrowing cost for entrepreneurs. Need to double check the current version of code, why the entrepreneurship changes with level of asset holding in the absence of credit friction.

6.6 Job market experiment

Show that necessity entrep rises as job market weakens via a) fall in match effciency, b) fall in productivity entrep tallent

Table 6: Impact of consumption floor on entrepreneurship, UI=0.0, lower matchig efficiency/higher search cost

	Baseline, UI=0.0	Lump-sum transfer	Penalty
entrepreneur-avg			
entrepreneur-low			
entrepreneur-mid			
entrepreneur-high			
firm size-avg			
firm size-low			
firm size-mid			
firm size-high			
Unemp-Ent			

Table 7: Impact of consumption floor on entrepreneurship, UI=0.6

	Baseline, UI=0.6	Lump-sum transfer	Penalty
entrepreneur-avg	0.144	0.146	0.145
entrepreneur-low	0.135	0.136	0.140
entrepreneur-mid	0.141	0.142	0.141
entrepreneur-high	0.154	0.156	0.155
firm size-avg	21.177	21.155	21.115
firm size-low	9.814	9.817	9.539
firm size-mid	17.023	17.023	17.056
firm size-high	32.958	32.885	32.793
Unemp-Ent	0.00084	0.0011	0.00117

Table 8: Impact of consumption floor on entrepreneurship, w/o unemployment

	no unemployment	Lump-sum transfer	Penalty
entrepreneur-avg	0.197	0.197	0.197
entrepreneur-low	0.207	0.207	0.206
entrepreneur-mid	0.184	0.184	0.184
entrepreneur-high	0.216	0.216	0.216
firm size-avg	24.126	24.127	24.131
firm size-low	9.884	9.884	9.903
firm size-mid	19.814	19.815	19.817
firm size-high	37.214	37.216	37.219
Unemp-Ent			

7 Conclusion

Appendix

Computation

Given the values for parameters, the discretized spaces Ω_a for a, Ω_z for z and Ω_x for x. Each agent draw a human capital level h (interpreted as educational attainment) at the beginning of the economy, which will be fixed henceforth. h will determine their labor productivity $z \in \Omega_z$, the separation rate ρ, and managerial ability $x \in \Omega_x$. Let $i_w = 0$ indicating being entrepreneur, $i_w = 1$ being employed and $i_w = 2$ being unemployed. The numerical algorithm works as follows.

1. Set a tolerance $\epsilon > 0$.

2. Guess $\Phi^0 = \left(r^0, w^0, \rho^0, \tau_y^0\right)$. Solve for optimal household behavior:

$$f : (\varphi; \Phi) \to \left(c, a', \mathcal{I}_e, n, k\right),$$

where $\varphi = \{h, z, x, a, i_w\}$. We will use the method of value function iteration as follows.

   (a) Guess value function $\mathbf{V}^0\left(\varphi; \Phi^0\right)$ and policy functions $f^0\left(\varphi; \Phi^0\right)$.

   (b) Calculate the cutoff values of x_u^* (for previously unemployed worker, or entrepreneur) and x_n^* (for previously employed worker) via the following conditions.

$$\widetilde{w}_u = \pi(x_u^*) = \max_n \left\{ (x_u^*)^{1-\alpha-\gamma} k^\alpha n^\gamma - w(1+\psi\tau)n - rk - \frac{\kappa z n}{q(\theta)} \right\} \tag{11}$$

$$\widetilde{w}_n = \pi(x_n^*) = \max_n \left\{ (x_n^*)^{1-\alpha-\gamma} k^\alpha n^\gamma - w(1+\psi\tau)n - rk - \frac{\kappa z n}{q(\theta)} \right\} \tag{12}$$

where $\widetilde{w}_n$ and $\widetilde{w}_u$ denote the expected compensation for the employed worker and unemployed worker respectively. More specifically,

$$\widetilde{w}_n = (1-\rho)\left\{zw(1-(1-\psi)\tau)\right\} + \rho b \tag{13}$$

$$\widetilde{w}_u = p(\theta)\left\{zw(1-(1-\psi)\tau)\right\} p(\theta))b. \tag{14}$$

Noting that the separation rate ρ is pre-determined by the individual's initial human capital level h.

(c) Update value and policy functions:

$$\mathbf{V}^1(h,z,x,a,i_w) = \tag{15}$$
$$\max_{\{a',c,\mathcal{I}_e\}} \left\{ \begin{array}{l} \iota_{i_w=0} \cdot \left[\begin{array}{c} (1-\mathcal{I}_e) \cdot \mathbb{E}_{p(\theta)} \left(V_w + \beta \mathbb{E} \mathbf{V}^0(h,z',x',a',i_w') \right) \\ +\mathcal{I}_e \cdot \left(V_e + \beta \mathbb{E} \mathbf{V}^0(h,z',x',a',i_w'=2) \right) \end{array} \right] + \\ \iota_{i_w=1} \cdot \left[\begin{array}{c} (1-\mathcal{I}_e) \cdot \mathbb{E}_{\rho} \left(V_w + \beta \mathbb{E} \mathbf{V}^0(h,z',x',a',i_w') \right) \\ +\mathcal{I}_e \cdot \left(V_e + \beta \mathbb{E} \mathbf{V}^0(h,z',x',a',i_w'=2) \right) \end{array} \right] + \\ \iota_{i_w=2} \cdot \left[\begin{array}{c} (1-\mathcal{I}_e) \cdot \mathbb{E}_{p(\theta)} \left(V_w + \beta \mathbb{E} \mathbf{V}^0(h,z',x',a',i_w') \right) \\ +\mathcal{I}_e \cdot \left(V_e + \beta \mathbb{E} \mathbf{V}^0(h,z',x',a',i_w'=2) \right) \end{array} \right] \end{array} \right\}$$
$$f^1\left(\varphi;\Phi^0\right) = \arg\max \mathbf{V}^1\left(\varphi;\Phi^0\right)$$

where $\mathbb{E}_{p(\theta)}\left(V_w + \beta \mathbb{E} \mathbf{V}(a',x',i_w')\right)$ denotes the expected payoff for the individual who is previously unemployed worker, or entrepreneur, and decides to look for paid job, while $\mathbb{E}_{\rho}\left(V_w + \beta \mathbb{E} \mathbf{V}(a',x',i_w')\right)$ represents the expected payoff for the previously employed worker who decide to stay in workplace.

$$\begin{aligned} \mathbb{E}_{p(\theta)}\left(V_w + \beta \mathbb{E} \mathbf{V}(h,z',x',a',i_w')\right) &= p(\theta)\left(V_n + \beta \mathbb{E} \mathbf{V}(h,z',x',a',i_w'=1)\right) \qquad (16) \\ &\quad + (1-p(\theta))\left(V_u + \beta \mathbb{E} \mathbf{V}(h,z',x',a',i_w'=0)\right) \\ \mathbb{E}_{\rho}\left(V_w + \beta \mathbb{E} \mathbf{V}(h,z',x',a',i_w')\right) &= (1-\rho)\left(V_n + \beta \mathbb{E} \mathbf{V}(h,z',x',a',i_w'=1)\right) \qquad (17) \\ &\quad + \rho\left(V_u + \beta \mathbb{E} \mathbf{V}(h,z',x',a',i_w'=0)\right) \end{aligned}$$

To simplify the analysis, we assume that the individual who was previously employed and newly separated from the firm is not allowed to search for job at the current period. The measure for the individuals who are looking for jobs, u_t, the total vacancies v_t, and job matching rate $\theta_{n,t}$ are given as follows.

$$u_t = \left(\int_x \iota_{i_w=0} + \int_x \iota_{i_w=2} \right) \cdot \int_0^{x_u^*} \tag{18}$$
$$v_t = \int_x \iota_{i_w=1} \cdot \rho \tag{19}$$
$$\theta_{n,t} = \bar{m} \left(\frac{v_t}{u_t} \right)^{1-\mu} \tag{20}$$

Note that we can write the above value functions explicitly as a system of subproblems: $P^{i_w=0}_{x<x_u^*}$, $P^{i_w=0}_{x\geq x_u^*}$, $P^{i_w=1}_{x<x_n^*}$, $P^{i_w=1}_{x\geq x_n^*}$, $P^{i_w=2}_{x<x_u^*}$, $P^{i_w=2}_{x\geq x_u^*}$, which correspond to the dynamic programing problem for i). previously self-employed with $x < x_u^*$; ii). previously self-employed with $x \geq x_u^*$; iii). previously employed with $x < x_n^*$; iv). previously employed with $x \geq x_n^*$; v). previously unemployed with $x < x_u^*$; vi). previously unemployed with $h \geq h_u^*$. To economize the space, we only provide the details for

$P^{i_w=0}_{x<x^*_u}$, and $P^{i_w=0}_{x\geq x^*_u}$ as follows.
$\left(P^{i_w=0}_{x<x^*_u}\right)$:

$$\mathbf{V}^1(h,z,x<x^*_u,a,i_w=0) = \max_{a'_n,a'_u}\left\{\begin{array}{c} p(\theta)\left(u(c_n)+\beta\mathbb{E}\mathbf{V}^0(h,z',x',a'_n,i'_w=1)\right) \\ +(1-p(\theta))\left(u(c_u)+\beta\mathbb{E}\mathbf{V}^0(h,z',x',a'_u,i'_w=0)\right)\end{array}\right\} \tag{21}$$

s.t.

$$c_n + a'_n \leq (1-\tilde{r}-\delta)a + zw(1-(1-\psi)\tau), \text{if matched with a firm} \tag{22}$$

$$c_u + a'_u \leq (1-\tilde{r}-\delta)a + b, \text{if unemployed} \tag{23}$$

$\left(P^{i_w=0}_{x\geq x^*_u}\right)$:

$$\mathbf{V}^1(h,z,x\geq x^*_u,a,i_w=0) = \max_{a'_e}\left\{u(c_e)+\beta\mathbb{E}\mathbf{V}^0(h,z',x',a'_e,i'_w=w)\right\} \tag{24}$$

s.t.

$$c_e + a'_e \leq (1-\tilde{r}-\delta)a + \nu(k,x;\tilde{r},w,\tau) \tag{25}$$

where

$$\nu(k,x;\tilde{r},w,\tau) = \max_n\left\{x^{1-\alpha-\gamma}k^\alpha n^\gamma - w(1+\psi\tau)n - rk - \frac{\kappa\rho n}{q(\theta)}\right\}. \tag{26}$$

(d) Stop if $\max\left\{\left|\mathbf{V}^1-\mathbf{V}^0\right|,\left|f^1-f^0\right|\right\}\leq\epsilon$. Set $\mathbf{V}^*=\mathbf{V}^1$, and $f^*=f^1$. Otherwise, set $\mathbf{V}^0=\mathbf{V}^1$, $f^0=f^1$ and repeat from step (c).

3. Generate a large number of individuals, $N=100000$. For each agent j assign a vector of initial condition $\left(h^j_t, z^j_0, x^j_0, a^j_0, i^j_{w,0}\right)$, where $x^j_0\sim\Omega_x$, $z^j_0\in\Omega_z$, $i^j_{w,0}\in\Omega_w$.

4. Simulate the economy for T periods, where T is sufficiently large.

5. Calculate the following statistics from the simulated path $\left\{h_t^j, z_t^j, x_t^j, a_t^j, i_{w,t}^j, \mathcal{I}_{e,t}^j, n_t^j, k_t^j\right\}_{t=0}^T$.

$$LS^0 = \frac{\sum_{j=1}^N \left(\iota_{i_w=1} \cdot z^j\right)}{N} \tag{27}$$

$$KS^0 = \frac{\sum_{j=1}^N \left(\iota_{i_w=0} \cdot a^j + \iota_{i_w=1} \cdot a^j + \iota_{i_w=2} \cdot a^j - \iota_{i_w=0} \cdot k^j\right)}{\sum_{j=1}^N a^j} \tag{28}$$

$$u_t = \left(\int_h \iota_{i_w=0} + \int_h \iota_{i_w=2}\right) \cdot \int_0^{h_u^*} \tag{29}$$

$$v_t = \int_h \iota_{i_w=1} \cdot \rho \tag{30}$$

$$p(\theta)_t = \bar{m}\left(\frac{v_t}{u_t}\right)^{1-\mu} \tag{31}$$

$$G = \int \tau w z \tag{32}$$

and τ_y^1 that balances the government's budget.

6. Stop and set $(r^*, w^*, \theta^*) = \left(r^0, w^0, \theta^0\right)$, if $\max\left\{LS^0, KS^0, \left|\theta_n^1 - \theta_n^0\right|\right\} \leq \epsilon$. Otherwise, update aggregate variables (restart from step 2):

$$\begin{aligned} r^0 &= \chi r^0 + (1-\chi) r^1 \left(KS^0\right) \\ w^0 &= \chi w^0 + (1-\chi) w^1 \left(LS^0\right) \\ p(\theta)^0 &= \chi p(\theta)^0 + (1-\chi) p(\theta)^1 \\ \tau_y^0 &= \chi \tau_y^0 + (1-\chi)\tau_y^1 \end{aligned}$$

where $\chi \in (0,1)$ is the step for updating aggregate variables.

Appendix Search and Matching

I am not sure how much of this is needed for coding purposes but this is the standard search and matching model. The stock of employed workers n_t can be expressed as the current pool of workers who did not separate from firms, plus new hires

$$n_t = (1-\rho) n_{t-1} + m_{t-1}$$

The matching function is Cobb-Douglass where

$$m_t = \bar{m} v_t^{1-\mu} u_t^{\mu}$$

$\bar{m}$ is a match efficiency parameter , v is vacancies, u is unemployment, and match elasticity is $0 < \mu < 1$.The hiring rate can be defined for firms as total number of new hires to the number of vacancies

$$q(\theta_t) = \frac{m_t}{v_t} = \bar{m}\theta_t^{-\mu}$$

where $\theta_t = \frac{v_t}{u_t}$is aggregate labour market tightness. Similarly we can define this from the perspective of the worker in the job-finding rate.

$$p(\theta_t) = \frac{m_t}{u_t} = \bar{m}\theta_t^{1-\mu}$$

The Job Creation Condition determines the amount of new vacancies an employer will post. In our model, most workers are on rolling contracts, so firms only need to post new vacancies after some workers separate from these rolling contracts. Usually in search and matching models the wage is determined ex-post of matching by Nash bargaining between the employer and employee. However, in our span-of-control model the wage has already been determined ex ante, as the wage is an endogenous variable that determines occupational choice. The marginal benefit that the new employee would bring to the firm is expressed as A (which can be derived from the firms production function, $\gamma X k^{\alpha}(zn)^{\gamma-1}$). The JCC can be then expressed as the costs of replacing out workers on the LHS

$$\frac{\kappa}{q(\theta_t)} = A - w_t + \beta E_t\left[A - w_{t+1} + (1-\rho)\frac{\kappa}{q(\theta_{t+1})}\right]$$

where κ is the cost of posting a vacancy.

Matching Steady States

To find the steady state of the employment equation $n_t = (1-\rho)n_{t-1} + m_{t-1}$, we use the fact that $u = (1-n)$ and $m_t = \bar{m}v_t^{1-\mu}u_t^{\mu}$. Substituting and getting rid of time subscript gives $(1-u) = (1-\rho)(1-u) + \bar{m}v^{1-\mu}u^{\mu}$ which simplifies to $\bar{m}v_t^{1-\mu}u_t^{\mu}$ $=\rho(1-u)$. Making v the subject of the foruma gives us the Beveridge curve.

$$v = \left(\frac{\rho(1-u)}{\bar{m}u}\right)^{1-\mu} u$$

The steady state for the JCC can be found by using $\frac{\kappa}{q(\theta_t)} = A - w_t + \beta E_t\left[A - w_{t+1} + (1-\rho)\frac{\kappa}{q(\theta_{t+1})}\right]$ and $q(\theta_t) = \bar{m}\theta_t^{-\mu} = \bar{m}\left(\frac{v_t}{u_t}\right)^{-\mu}$. Substituting and getting rid of time subscripts yields

$$\frac{\kappa}{\bar{m}}\left(\frac{v}{u}\right)^{\mu} = A - w_t + \beta\left[A - w + (1-\rho)\frac{\kappa}{\bar{m}}\left(\frac{v}{u}\right)^{\mu}\right]$$

Simplifying yields the steady state of the JCC

$$1 - \beta(1-\rho)\frac{\kappa}{\bar{m}}\left(\frac{v}{u}\right)^{\mu} = (1+\beta)\left[A - w\right]$$